

Excellent!

$\frac{9}{9} \Rightarrow 100\%$

0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	100
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Name: _____

These questions assess your understanding of the material from Chapters 16, 17 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Suppose a train sounding its 678.-Hz horn is moving at $31.0 \frac{m}{s}$. What frequency does a person standing by the tracks hear after the train passes? Given this particular day's temperature, the speed of sound is $329. \frac{m}{s}$.

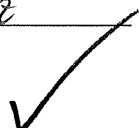
678 Hz
31.0 $\frac{m}{s}$
329 $\frac{m}{s}$
passes $\rightarrow +$

$$f_{obs} = f_s \left(\frac{v_w}{v_w \pm v_s} \right)$$

$$f = (678) \left(\frac{329}{329 + 31} \right)$$

$$f = 619.6 \text{ Hz}$$

ANSWER: 619.6 Hz



2. Calculate the sound intensity level in decibels for a sound wave traveling in air at 2.2°C with a pressure amplitude of 2.30 Pa. Recall that the density of air is $1.29 \frac{kg}{m^3}$ and the speed of sound in air at 0°C is $331 \frac{m}{s}$.

$2.2^\circ\text{C} \rightarrow 275.2 \text{ K}$
 $\Delta p = 2.30 \text{ Pa}$
 $\rho = 1.29 \frac{kg}{m^3}$

$$\textcircled{1} v_w = (331) \left(\sqrt{\frac{T}{273}} \right)$$

$$v_w = (331) \sqrt{\frac{275.2}{273}}$$

$$v_w = 332.3 \text{ m/s}$$

$$\textcircled{2} I = \frac{(\Delta p)^2}{2 \rho v_w}$$

$$I = \frac{(2.3)^2}{2(1.29)(332.3)}$$

$$I = 6.17 \times 10^{-3} \frac{W}{m^2}$$

ANSWER: 97.9 dB

$$\textcircled{3} B(\text{dB}) = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

$$B = 10 \log_{10} \left(\frac{6.17 \times 10^{-3}}{1 \times 10^{-12}} \right)$$

$$B = 97.9 \text{ dB}$$

3. Ultrasound is transmitted from fat to muscle during a particular medical procedure. Calculate the percentage of the ultrasound's intensity that is reflected when the ultrasound travels from fat with a density of $919. \frac{kg}{m^3}$ to muscle tissue with a density of $1.165 \times 10^3 \frac{kg}{m^3}$. The speed of ultrasound in the fat is $1.47 \times 10^3 \frac{m}{s}$ and the speed of ultrasound in the muscle tissue is $1.53 \times 10^3 \frac{m}{s}$.

fat $\rightarrow$ muscle

$\rho_f = 919 \frac{kg}{m^3}$
 $v_f = 1.47 \times 10^3 \frac{m}{s}$
 $\rho_m = 1.165 \times 10^3 \frac{kg}{m^3}$
 $v_m = 1.53 \times 10^3 \frac{m}{s}$

$$\textcircled{1} Z = \rho v$$

fat (1) muscle (2)

$$Z_f = (919)(1.47 \times 10^3) \quad Z_m = (1.165 \times 10^3)(1.53 \times 10^3)$$

$$Z_f = 1.35 \times 10^6 \quad Z_m = 1.78 \times 10^6$$

$$\textcircled{2} a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2}$$

$$a = \frac{((1.78 \times 10^6) - (1.35 \times 10^6))^2}{((1.35 \times 10^6) + (1.78 \times 10^6))^2}$$

$$a = 1.90 \times 10^{-2}$$

ANSWER: 1.90%

$$\textcircled{3} a \Rightarrow \% = a(100)$$

$$\% = (1.90 \times 10^{-2})(100)$$

$$a\% = 1.90\%$$

+3

4. Suppose you have a tube that is closed at one end. Its length is 11 cm and the air temperature is 28°C. What is the frequency of its third overtone?

$$11 \text{ cm} \rightarrow 11 \times 10^{-2} \text{ m}$$

$28^{\circ}\text{C} \rightarrow 301\text{ K}$

3rd overtone

$$h=7$$

$$\textcircled{1} \quad V_w = (331) \sqrt{\frac{T}{273}}$$

$$V_w = (331) \sqrt{\frac{301}{273}}$$

$$V_w = 347.6 \frac{\text{m}}{\text{s}}$$

ANSWER: $5.53 \times 10^3 \text{ Hz}$

$$(2) f_n = n \frac{v_w}{4L}$$

$$f_2(\gamma) = \frac{347.6}{4 \times 10^{-2}}$$

$$F = 5.53 \times 10^7 \text{ Hz}$$

5. When William gets in his car, it lowers by 1.2 cm. If William's mass is 85 kg, then what is the force constant of his car's suspension system?

$$-1.2 \text{ cm} \rightarrow -1.2 \times 10^{-2} \text{ m}$$

85 kg

$$\begin{aligned} \textcircled{1} \quad F &= ma \\ F &= (85)(9.8) \\ F &= 833 \text{ N} \end{aligned}$$

(2) $F = -k \Delta x$

$$833 = -k(-1.2 \times 10^{-7})$$

$$k = 6.94 \times 10^4 \frac{N}{m}$$

ANSWER: $6.94 \times 10^4 \frac{\text{kg}}{\text{m}^3}$

6. Suppose that a 1.11×10^3 -kg car has a suspension system with a force constant $2.98 \times 10^4 \frac{\text{N}}{\text{m}}$. If the car hits a bump and bounces with an amplitude of 5.5 cm, then what is its maximum vertical velocity assuming no damping occurs?

$$1.11 \times 10^3 \text{ kg}$$

$$2.98 \times 10^4 \frac{\text{g}}{\text{m}^3}$$

$$\lambda = 5.5 \text{ cm} \rightarrow 5.5 \times 10^{-2} \text{ m}$$

$$\textcircled{1} \quad v(t) = \sqrt{\frac{k}{m}} \times \underbrace{\sin \frac{2\pi t}{T}}_{\sin = 1} \quad \text{max}$$

$$v(t) = \sqrt{\frac{2.98 \times 10^4}{1.1 \times 10^3}} (5.5 \times 10^{-2}) \text{ m/s}$$

$$v(t) = 0.285 \text{ m/s}$$

ANSWER: 0.285 $\frac{m}{s}$

7. If the shock absorbers in a car go bad, then the car will oscillate at the smallest provocation, such as when going over bumps in the road and after stopping. Calculate the period of oscillations for a car with shoddy shock absorbers if the car's total mass is $1.04 \times 10^3 \text{ kg}$ and the force constant of the suspension system is $5.75 \times 10^4 \frac{\text{N}}{\text{m}}$.

$$1.04 \times 10^3 \text{ kg}$$

$$5.75 \times 10^4 \frac{\text{m}}{\text{s}}$$

$$\textcircled{1} T_{540} = 2\pi \sqrt{\frac{m}{K}}$$
$$T = 2\pi \sqrt{\frac{1.04 \times 10^{-3}}{5.75 \times 10^4}}$$
$$T = 0.845 \text{ s}$$

ANSWER: 0.845 s

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8. A ukulele makes a note with a frequency of 384. Hz. How long does it take the string to complete one oscillation?

$$384 \text{ Hz}$$

$$\textcircled{1} f = \frac{1}{T}$$

$$384 = \frac{1}{T}$$

$$T = \frac{1}{384}$$

$$T = 2.60 \times 10^{-3} \text{ s}$$

ANSWER: $2.60 \times 10^{-3} \text{ s}$



9. How much energy is stored in the spring of a toy gun that has a force constant of $84. \frac{\text{N}}{\text{m}}$ and is compressed 19. cm?

$$84 \frac{\text{N}}{\text{m}}$$

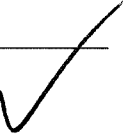
$$19 \text{ cm} \rightarrow 19 \times 10^{-2} \text{ m}$$

$$\textcircled{1} PE_{\text{el}} = \frac{1}{2} k x^2$$

$$PE = \frac{1}{2} (84) (19 \times 10^{-2})^2$$

$$PE = 1.52 \text{ J}$$

ANSWER: 1.52 J



+2

1. $f = 619.6 \text{ Hz} \therefore f_{obs} = f_s \left(\frac{v_w}{v_w \pm v_s} \right)$ (review the Doppler effect)
2. $\beta = 97.9 \text{ dB} \therefore v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}$, $I = \frac{(\Delta p)^2}{2\rho v_w}$, $\beta \text{ (dB)} = 10 \log_{10} \left(\frac{I}{I_0} \right)$, $I_0 = 1 \times 10^{-12} \frac{\text{W}}{\text{m}^2}$
3. $a = 1.90\% \therefore Z = \rho v$, $a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2}$
4. $f = 5.53 \times 10^3 \text{ Hz} \therefore f_n = n \frac{v_w}{4L}$, $v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}$
5. $k = 6.94 \times 10^4 \frac{\text{N}}{\text{m}} \therefore \vec{F} = -k\Delta\vec{x}$. $k = -\frac{\vec{F}}{\Delta\vec{x}}$
6. $v_{max} = 0.285 \frac{\text{m}}{\text{s}} \therefore v(t) = \sqrt{\frac{k}{m}} X \sin \frac{2\pi t}{T}$ and the max of sine's range is 1.
7. $f = 0.845 \text{ s} \therefore T_{SHO} = 2\pi \sqrt{\frac{m}{k}}$
8. $T = 2.60 \text{ ms} \therefore T = \frac{1}{f}$
9. $PE = 1.52 \text{ J} \therefore PE = \frac{1}{2} kx^2$